

The Illusion of Loyalty Behind Discounts

Power BI Project | Customer Insight & Discount Effectiveness Analysis

Summary

This project examines whether discount-driven purchasing actually reflects customer loyalty. Using a Power BI dashboard, the analysis compares discount and non-discount customers across spend, items purchased, satisfaction, and repurchase timing. The findings show that discounts can lift short-term transaction value, but they do not consistently translate into stronger repeat-purchase behavior—suggesting that discount response and loyalty are not the same thing.

Project Overview

The objective of this analysis is to evaluate whether customers who buy with discounts display stronger loyalty than those who purchase at regular price. The dashboard is structured to compare both groups by gender and age range, enabling a clearer view of how promotions influence spending behavior and customer retention.

Dataset & Tools

Dataset: Customer-level purchase and satisfaction data
Primary Tool: Microsoft Power BI
Techniques: Segmentation analysis, comparative KPI analysis, scatter plot interpretation, dashboard storytelling
Dimensions: Discount status, gender, age range, spend, item count, rating, days since last purchase

Key KPI Snapshot

350 Total Customers	845.38 Avg Total Spend	12.60 Avg Items Purchased
4.02 Avg Rating	26.59 Avg Days Since Last Purchase	50.00% Discount Customer %

Power BI Dashboard

Dashboard view used to compare discount and non-discount customers across key behavioral metrics.

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Avg Rating

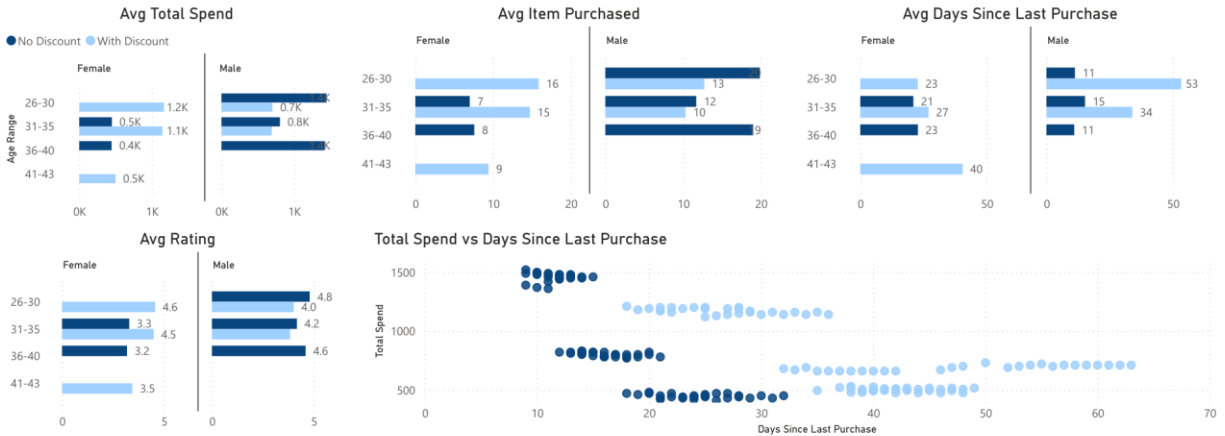
26.59

Avg Days Since Last Purchase

50.00%

Discount Customer %

Discount vs. Non-discount Comparison



Key Insights

1. Discounted customers often generate higher basket value and item volume in several demographic groups, especially among customers aged 26–35. This indicates that promotions can successfully stimulate short-term purchasing activity.
2. Repurchase timing is not consistently better for discount-driven customers. In several segments, customers who bought with discounts show longer gaps since their last purchase, which weakens the assumption that discount usage equals loyalty.
3. Customer ratings remain relatively solid across both groups, meaning the main difference lies less in satisfaction and more in purchasing behavior. In other words, promotions can influence transaction size without guaranteeing habit formation.
4. The scatter plot between total spend and days since last purchase reinforces the central story: higher spend can happen alongside weaker recency, suggesting that some customers respond to offers opportunistically rather than loyally.

Business Recommendations

1. Refocus promotions from broad discounting to targeted offers aimed at segments with the highest conversion potential, especially where discounts lift spend but repeat behavior remains weak.
2. Build loyalty mechanisms beyond price cuts, such as points, personalized offers, member-only benefits, or replenishment reminders, to encourage repeat purchases without overtraining customers to wait for discounts.
3. Track retention KPIs separately from campaign KPIs. Revenue uplift during promotions should be monitored alongside repeat-purchase interval, retention rate, and customer lifetime value so that promotional success is not judged only by immediate sales.

4. Use segmentation in future campaigns. Since behavior differs by age range and gender, discount strategy should be tested by audience segment instead of applying one uniform promotion model to the entire customer base.

Analytical Focus

The dashboard evaluates loyalty through four complementary lenses so the interpretation goes beyond sales uplift alone.

Metric	What It Reveals
Avg Total Spend	Measures whether discounts increase order value and how strongly promotions drive basket expansion.
Avg Items Purchased	Shows whether promoted customers add more products per transaction or only react on price-sensitive items.
Avg Rating	Helps separate satisfaction from purchasing behavior, indicating whether higher spend is matched by a better experience.
Days Since Last Purchase	Acts as a loyalty signal by showing whether customers come back sooner or disappear after taking the offer.

Business Impact This analysis helps decision-makers distinguish transactional response from real loyalty. By separating short-term promotion lift from repeat-purchase behavior, the business can improve pricing strategy, reduce margin erosion, and design retention initiatives that create healthier long-term customer value.	Conclusion The project demonstrates how Power BI can transform customer-level data into a strategic narrative. The main takeaway is clear: discounts may increase spend, but they do not automatically build loyalty. Stronger retention requires a broader customer strategy than discounting alone.
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